

A machine-verified erratum and its repair: Brown–Zudilin’s $\zeta(5)$ denominators and the Frobenius certificate, formalized in Lean 4

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Abstract

Brown and Zudilin’s paper on cellular rational approximations to $\zeta(5)$ [1] states, as an experimental observation, the integrality $d_n^5 P_n \in \mathbb{Z}$ for the rational coefficients of its totally symmetric linear forms, where $d_n = \text{lcm}(1, \dots, n)$. The paper’s own displayed value $P_2 = 1190161/384$ contradicts this inclusion, failing by a factor of exactly 12; the claim is still present, unamended, in the January 2026 revision. We present a Lean 4 formalization, mechanically verified by the Lean kernel with zero `sorry`s and only the three standard axioms, of: (i) the contradiction, with P_2 derived from the construction rather than transcribed; (ii) its sharp repair at $n = 2$: $c d_2^5 P_2 \in \mathbb{Z}$ if and only if $12 \mid c$ (companion form: if and only if $2 \mid c$); and (iii) the theorem that proves the corrected law on the clean interval — the *Frobenius certificate* theorem of [3]: for every $n \geq 1$ and every prime p with $5 \leq p$, $n < p \leq 2n$, the harmonic functionals satisfy $w_n \equiv \tilde{w}_n \equiv 0 \pmod{p}$, and hence $p \mid p_n$. The formalization exposed two statement-level errors in our own paper-level drafts, which we document. The complete development builds in one `lake build` and is publicly available.

1 Introduction

The irrationality theory of odd zeta values rests on rational linear forms in zeta values whose coefficients must be cleared of denominators at a controlled exponential cost: for a weight-5 form the benchmark cost is d_n^5 , with $d_n = \text{lcm}(1, 2, \dots, n) \sim e^n$. In the totally symmetric cellular construction of Brown and Zudilin [1], the linear form

$$I_n = Q_n(2\zeta(5) + 4\zeta(3)\zeta(2)) - 4\hat{P}_n \zeta(2) - 2P_n \quad (1)$$

carries rational coefficient ladders Q_n , $\hat{P}_n$, P_n , and the paper states (in the display labeled `dn-totsym`, presented explicitly as an experimental observation from “an extensive computation of the rational coefficients”):

$$Q_n, d_n^2 d_{2n} \hat{P}_n, d_n^5 P_n \in \mathbb{Z} \quad (n = 0, 1, 2, \dots). \quad (2)$$

The same paper displays (in the block following the definition of I_n) the exact values

$$Q_2 = 2989, \quad \hat{P}_2 = \frac{344923}{96}, \quad P_2 = \frac{1190161}{384}. \quad (3)$$

Since $d_2 = 2$ and $384 = 2^7 \cdot 3$, we have $d_2^5 P_2 = 1190161/12 \notin \mathbb{Z}$: the third inclusion of (2) fails already at $n = 2$, by a factor of exactly 12 (and the second fails by a factor of exactly 2: $d_2^2 d_4 \hat{P}_2 = 344923/2$). The corrected law

$$12 d_n^5 P_n \in \mathbb{Z}, \quad (4)$$

conjecturally with the $\{2, 3\}$ -ceilings sharp, has substantial computational support, and its clean-interval part is a theorem: the *Frobenius certificate* theorem of the companion paper [3] shows that every prime $n < p \leq 2n$ (the primes of $\binom{2n}{n}$, which the naïve partial-fraction bound admits into the denominator) in fact divides the numerator p_n .

This note reports a complete formalization of the erratum, its sharp $n = 2$ repair, and the clean-interval theorem in Lean 4 over Mathlib. Every theorem below is verified by the Lean kernel; the development contains zero `sorry`s, uses no `native_decide`, adds no axioms, and each named theorem reports exactly the standard axiom triple `[propext, Classical.choice, Quot.sound]`. The trust base is the Lean kernel and the printed displays of [1] quoted above — nothing else.

Two purposes motivate machine verification here. First, an erratum to a published paper should meet a higher evidentiary standard than the claim it corrects; a kernel-checked derivation of P_2 from the construction, ending in a formal negation of (2) at $n = 2$, is the strongest form of such a statement we know how to produce. Second, the formalization is a test of a design thesis: proofs shaped as *finite certificates plus one small uniform identity* formalize at low cost. The clean-interval theorem has exactly this shape, and the entire development — definitions, numeric gates, five lemma modules, assembly — was completed in a single day. As a byproduct the formalization caught two statement-level errors in our own drafts (Section 5), neither of which affects the mathematics, both of which would likely have survived human refereeing.

2 The construction and the formal objects

Fix $n \geq 1$ and set

$$R_n(k) = (n!)^4 \left(k + \frac{n}{2}\right) \frac{\prod_{j=1}^n (k-j) \prod_{j=1}^n (k+n+j)}{\prod_{j=0}^n (k+j)^6}, \quad (5)$$

Zudilin’s totally symmetric rational function [2]. Its partial fractions

$$R_n(k) = \sum_{i=1}^6 \sum_{j=0}^n \frac{a_{i,j}}{(k+j)^i}, \quad \tilde{a}_{i,j} = j(n-j)a_{i,j} + (2j-n)a_{i+1,j} - a_{i+2,j}$$

(indices above 6 read as zero; the $\tilde{a}$ are the partial fractions of $-k(k+n)R_n(k)$) define the functionals

$$u = \sum_j a_{5,j}, \quad w = \sum_j a_{3,j}, \quad v = \sum_{i,j} a_{i,j} H_j^{(i)}, \quad H_j^{(i)} = \sum_{m=1}^j m^{-i},$$

their tilded companions, and the determinant ladders

$$q_n = u\tilde{w} - \tilde{u}w, \quad p_n = \tilde{w}v - w\tilde{v}, \quad \tilde{p}_n = u\tilde{v} - \tilde{u}v. \quad (6)$$

The Brown–Zudilin coefficients of (1) correspond via $P_n = (-1)^{n+1}p_n/\binom{2n}{n}$ (the display following (2) in [1], which ties the cellular normalization to [2]).

In Lean, all of these are *computable* functions on $\mathbb{Q}$: the primary definition of $a_{i,j}$ is the truncated local Taylor expansion of $B_j(k) = (k+j)^6 R_n(k)$ at $k = -j$ (module `Cbcert/Defs.lean`), and the partial-fraction property is then a *theorem* (`pf_cleared`, module `Cbcert/PartialFraction.lean`): the cleared polynomial identity

$$N_n(k) = \sum_{i,j} a_{i,j} (k+j)^{6-i} \prod_{m \neq j} (k+m)^6,$$

proven for all n . Computability is what makes the development self-auditing: a numeric gate (`Cbcert/Numeric.lean`) reduces the full definition chain by kernel computation at $n = 2$,

$$w = \frac{6125}{4}, \quad \tilde{w} = 1764, \quad u = 469, \quad \tilde{u} = 552, \quad v = \frac{74463}{32}, \quad \tilde{v} = \frac{43085}{16}, \quad p_2 = \tilde{w}v - w\tilde{v} = -\frac{1190161}{64},$$

and at $n = 3$, in exact agreement with the independent reference implementation; the frozen definitions were gated on this agreement before any lemma work began. Note $P_2 = (-1)^3 p_2 / \binom{4}{2} = 1190161/384$, matching (3) exactly.

3 The machine-verified erratum

Theorem 1 (`Cbcert/ErrorExhibit.lean`). *Define P_2 by the construction of Section 2 (derived, not transcribed). Then, verified by the Lean kernel with standard axioms:*

- (i) *`P2_eq`: $P_2 = 1190161/384$, and likewise $Q_2 = 2989$, $\hat{P}_2 = 344923/96$ — the construction reproduces all three displayed values (3) of [1].*
- (ii) *`bz_inclusion_fails`: $d_2^5 P_2 \notin \mathbb{Z}$. In particular the inclusion $d_n^5 P_n \in \mathbb{Z}$ of (2) is false at $n = 2$.*
- (iii) *`clearing_iff`: for every $c \in \mathbb{N}$, $c d_2^5 P_2 \in \mathbb{Z} \iff 12 \mid c$. The repair factor 12 of (4) is sharp at $n = 2$.*
- (iv) *`companion_inclusion_fails`, `companion_clearing_iff`: $d_2^2 d_4 \hat{P}_2 \notin \mathbb{Z}$, and $c d_2^2 d_4 \hat{P}_2 \in \mathbb{Z} \iff 2 \mid c$.*

The Lean statements are short enough to read in full (unicode transliterated):

```
theorem bz_inclusion_fails :
  ¬ ∃ m : ℤ, (d 2 : ℚ)^5 * P₂ = (m : ℚ)
theorem clearing_iff (c : ℕ) :
  (∃ m : ℤ, (c : ℚ) * (d 2)^5 * P₂ = m) ↔ 12 ∣ c
```

where `d` is the genuine running $\text{lcm}(1, \dots, n)$. Each definition and theorem carries a docstring citing the source labels of [1] (`dn-totsym` for the claim, the `I.n` display block for the values), so a reader can hold the PDF next to the Lean file. Two remarks on scope. The claim (2) is presented in [1] as an experimental observation, not a theorem; what is exhibited here is a false printed claim and its sharp correction, not a broken proof. And the bridge between the cellular normalization of [1] and the construction of Section 2 is the paper’s own stated correspondence to [2]; it is corroborated *inside* the formalization by Theorem 1(i), which reproduces all three displayed values simultaneously from one derivation chain.

4 The Frobenius-certificate theorem, formalized

The mathematical content, proven in [3]: for a prime p with $n < p \leq 2n$, the Frobenius identity $k^p - k \equiv (k + j)^p - (k + j) \pmod{p}$ makes $\varphi = (k^p - k)^2$ congruent to $(k + j)^2$ modulo $(k + j)^6$ at every pole $-j$, while $\deg \varphi = 2p \leq 4n + 2$ keeps φR_n decaying at infinity; the residue-sum theorem on $\mathbb{P}^1$ then forces $\sum_j a_{3,j} \equiv 0 \pmod{p}$. In certificate form this is the three-term relation with coefficients $(c_3, c_{p+2}, c_{2p+1}) = (1, -2, 1)$ among the decay relations

$$E_M = \sum_{i=1}^{\min(6,M)} \binom{-i}{M-i} \sum_j j^{M-i} a_{i,j} = 0 \quad (1 \leq M \leq 4n + 4),$$

which are the coefficients of the expansion of R_n at infinity.

Theorem 2 (Cbcert/Main.lean). *For every $n \geq 1$ and every prime p with $5 \leq p$ and $n < p \leq 2n$, verified by the Lean kernel with standard axioms:*

$$\text{res_congruence_w} : p \mid w_n, \quad \text{res_congruence_wt} : p \mid \tilde{w}_n, \quad \text{res_congruence_pn} : p \mid p_n,$$

where $p \mid x$ for a p -integral rational x means $\text{res}_p(x) = 0$ under the residue homomorphism res_p to $\mathbb{Z}/p$. Equivalently (*pn_valuation* etc.): $p_n = 0$ or $\text{ord}_p(p_n) \geq 1$.

The last statement of Theorem 2 is the clean-interval central-binomial cancellation (CB₁): every prime of the window $(n, 2n]$, $p \geq 5$, divides p_n .

Architecture

The proof formalizes in four independent modules plus an assembly, mirroring [3]:

module	content	status
Defs.lean	all definitions (computable over $\mathbb{Q}$)	frozen, gated
Certificate.lean	Lemma B: the $\mathbb{F}_p$ certificate identity	proven
PartialFraction.lean	pf_cleared: the all- n cleared identity	proven
Decay.lean	Lemma A: $E_M(a) = E_M(\tilde{a}) = 0$	proven
Integrality.lean	Lemma C: p -integrality of $a, \tilde{a}, H$	proven
Main.lean	residue map, reordering, main theorems	proven
Numeric.lean	kernel-reduction gates at $n = 2, 3$	proven
ErrorExhibit.lean	Theorem 1	proven

Lemma B is pure binomial arithmetic in $\mathbb{Z}/p$: for $p \geq 5$, $1 \leq i \leq 6$ and every $x \in \mathbb{Z}/p$,

$$\sum_{M \in \{3, p+2, 2p+1\}} c_M \binom{-i}{M-i} x^{M-i} = \delta_{i,3} \quad \text{in } \mathbb{Z}/p,$$

proven from $x^p = x$ and $p \mid \binom{p}{m}$ for $0 < m < p$. It is the mathematically novel part of [3] and was the *easiest* module to formalize — a data point for the certificate-shaped design thesis.

Lemma A never needed the anticipated Laurent-series machinery: with the cleared identity pf_cleared in hand, the decay relations become the statement that $E_M(a)$ is the coefficient of X^M in $R_n(1/X) = X^{4n+5} \tilde{N}(X)/\tilde{D}(X)$ with $\tilde{D}(0) = 1$, an elementary computation in $\mathbb{Q}[[X]]$. The load-bearing insight in pf_cleared itself was that the truncated-multiplication primitive used by the computable definitions *is* the Cauchy product, so the Taylor-coefficient definition of $a_{i,j}$ composes directly into the polynomial identity.

Lemma C bounds every denominator of $a_{i,j}$, $\tilde{a}_{i,j}$ ($j \leq n$) and $H_j^{(i)}$ by primes $\leq n < p$: the pole differences $|j - j'| \leq n$ are the only sources.

Assembly. A residue homomorphism res_p is developed on the p -integral subring of $\mathbb{Q}$ (compatible with $+$, $\times$, and with $\text{res}_p(x) = 0 \iff x = 0 \vee \text{ord}_p(x) \geq 1$), and a coefficient-agnostic reordering lemma turns Lemmas A + B + C into $\text{res}_p(\sum_j b_{3,j}) = 0$ for both coefficient systems $b \in \{a, \tilde{a}\}$ at once; $p \mid p_n$ then follows from (6) by ring arithmetic. All three certificate indices satisfy $M \leq 4n+4$ (respectively $4n+2$ for $\tilde{a}$) precisely because $p \leq 2n$; the case $p = 2n+1$ is genuinely excluded.

5 What the formalization caught

Two statement-level errors in our own paper-level drafts were discovered and repaired during formalization. We record them because both are invisible to ordinary numerical testing on the intended domain, and both concern the *statements*, not the proofs.

- (1) **The valuation phrasing is unsound at zero.** The intended statement $\text{ord}_p(p_n) \geq 1$ was originally frozen as $1 \leq \text{pAdicValRat } p (p_n)$. Mathlib defines $\text{pAdicValRat } p 0 = 0$, so this phrasing silently asserts $p_n \neq 0$ — a genuinely separate arithmetic fact (true numerically for $n \leq 60$; open in general, and *not* needed for any downstream denominator application). The canonical theorems are therefore stated in residue form, with the faithful disjunction $p_n = 0 \vee 1 \leq \text{pAdicValRat } p (p_n)$ provided as a corollary. General nonvanishing is documented as an explicitly open problem, not a hidden hypothesis.
- (2) **The integrality lemma needs its hypotheses.** As first staged, Lemma C quantified over all pole indices j and all primes $p > n$. Both restrictions matter: the formal coefficient function evaluated out of range or at $p = 2$ produces genuine counterexamples (at $n = 1$: $a_{5,0} = 9/2$, and the out-of-range $a_{5,2} = 9/128$ with $\text{ord}_2 = -7$). The proven lemma carries $j \leq n$ and p odd; the false general wrappers were deleted.

Neither finding affects [3], whose arguments always operate in range with $p \geq 5$; both are exactly the kind of boundary defect that survives refereeing.

6 Scope and open problems

The formalized results are: the erratum and its sharp $n = 2$ repair (Theorem 1), and the clean-interval theorem (Theorem 2) — the part of the corrected law (4) that is a theorem on paper. Explicitly *not* formalized, because they are not yet theorems on paper: the full corrected law for all n (its remaining gap is a Dwork-type descent congruence, for which extensive exact computational evidence exists, including a uniform mod p^5 jet on the $a = 1$ band); and the general nonvanishing $w_n, \tilde{w}_n, p_n \neq 0$. The finite-range law $12 d_n^5 p'_n \in \mathbb{Z}$ for $n \leq N$ by kernel computation is routine future work on top of the present definitions.

7 Verification

The development lives in the `cbcert/` directory of the public repository [6], pinned to Lean toolchain `v4.33.0-rc1` and a fixed Mathlib commit. To verify: `lake exe cache get && lake build` (the full build reports ≈ 8669 jobs and finishes with no sorry warnings), then `#print axioms` on any theorem named above; each reports exactly `[propext, Classical.choice, Quot.sound]`. The repository README contains the recipe, including which displays of [1] to compare against which Lean docstrings.

Provenance

The formalization was produced on 19 July 2026 in a single orchestrated session by a fleet of LLM agents (Claude, Anthropic), following the statement-first protocol described above: definitions frozen against exact numeric gates before lemma work, statements corrected only under the documented-erratum discipline of Section 5. As with any formal development, correctness rests on the Lean kernel and the pinned toolchain, not on the process that produced the proof scripts.

References

- [1] F. BROWN, W. ZUDILIN, On cellular rational approximations to $\zeta(5)$, arXiv:2210.03391 [math.NT] (2022; revised 26 January 2026).
- [2] W. ZUDILIN, A third-order Apéry-like recursion for $\zeta(5)$, *Mat. Zametki* **72** (2002), no. 5, 796–800; English transl., *Math. Notes* **72** (2002), no. 5–6, 733–737.
- [3] A Frobenius certificate for central-binomial cancellation in Zudilin’s symmetric $\zeta(5)$ construction, companion paper, this repository (`worthiness/cb_certificate.tex`), 17 July 2026; revised 19 July 2026.
- [4] L. DE MOURA, S. ULLRICH, The Lean 4 theorem prover and programming language, *CADE 28*, Lecture Notes in Comput. Sci. **12699** (2021), 625–635.
- [5] THE MATHLIB COMMUNITY, The Lean mathematical library, *CPP 2020*, ACM (2020), 367–381.
- [6] Odd zeta values: machine verifications (repository), <https://github.com/rain-1/odd-zeta-values-autoformalization>.